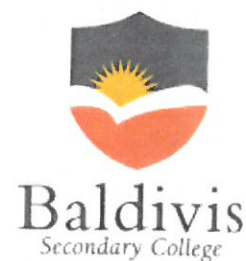


Year 10
Pathway 2
End of Term 1 Test
2018



Name: Solutions

Class: _____

Time: 55 minutes

Total Mark: ____/58

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 A4 page of notes (1 side), Calculator allowed.

Algebra

Question 1

2 marks

Simplify the following.

a) $4y - 5 + 7y + 1$

$11y - 4$ ✓

b) $4p - 3(2 - p)$

$4p - 6 - 3p$

$p - 6$ ✓

Question 2

4 marks

If $a = -4$, $b = 6$ and $c = 3$ calculate the following.

a) $a + b + c =$

$-4 + 6 + 3 = 5$ ✓

b) $3b - 2c + a =$

$18 - 6 - 4 = 8$ ✓

c) $(c - a)^2 =$

$(3 - (-4))^2 = (7)^2$
 $= 49$ ✓

d) $6a - 5c =$

$6(-4) - 5(3) = -24 - 15$
 $= -39$ ✓

Question 3

6 marks

Simplify the following indices. Give your answers as positive indices.

Simplify the following indices.

a) $2x^3 \cdot 3x^2y^2$

$6x^5y^2$ ✓

b) $\frac{a^3b^2}{a^2b^2} = a$ ✓

c) $m^2 \cdot n^3 \cdot n^4 \cdot m =$

m^3n^7 ✓

d) $\frac{12p^3q^2}{6pq^6} = \frac{2p^2}{q^4}$ ✓

e) $a^{-2} = \frac{1}{a^2}$ ✓

f) $\left(\frac{a^2}{2}\right)^2 = \frac{a^4}{4}$ ✓

Question 4

1, 2, 2, 1- 6 marks

Expand and simplify the following:

a) $2(2x - 3y + 5)$

$4x - 6y + 10$ ✓

c) $(x - 2)(x + 4)$

$x^2 - 2x + 4x - 8$ ✓
 $x^2 + 2x - 8$ ✓

b) $(2m + 2)(m - 4)$

$2m^2 - 8m + 2m - 8$ ✓
 $2m^2 - 6m - 8$ ✓

d) $-5(2a - 2b + c)$

$-10a + 10b - 5c$ ✓

Question 5**1, 1, 2, 2, 2 -8 marks**

Simplify the following algebraic fractions:

a) $\frac{4a}{2}$

$= 2a$

b) $\frac{2a}{10} + \frac{5a}{10} =$

$\frac{7a}{10}$

c) $\frac{9z}{10} - \frac{3z}{5} =$

$\frac{9z}{10} - \frac{6z}{10} = \frac{3z}{10}$

d) $\frac{6b}{3} \times \frac{4b}{10} =$

$\frac{24b^2}{30} = \frac{2b^2}{5}$

e) $\frac{5d}{6} \div \frac{5d}{8} =$

$\frac{5d}{6} \times \frac{8}{5d} = \frac{8}{6} = \frac{4}{3}$

Factorising**Question 6****3 marks****Factorise** the following: (Hint one bracket ())

a) $15c - 5$

$5(3c-1)$

b) $2x + 4x^2$

$2x(1+2x)$

c) $36ab + 12cb$

$12b(3a+c)$

Question 7**4 marks**Fully **factorise** each of the following. (Hint two brackets)

(a) $3(a-1) + b(a-1)$

$(a-1)(3+b)$

(b) $2p(3w+2) - 3(3w+2)$

$(3w+2)(2p-3)$

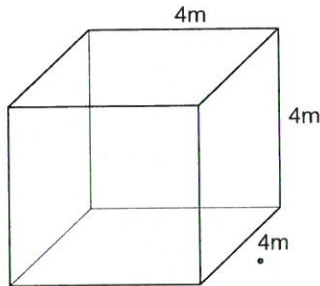
Surface Area

Question 9

2, 3, -5 marks

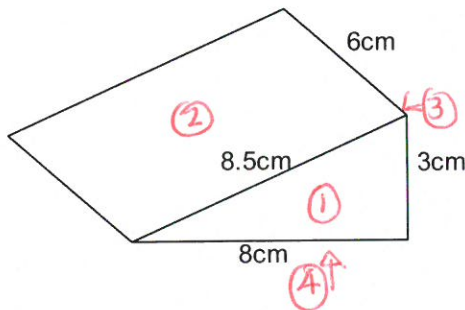
Determine the surface area of the following 3D Shapes:

a)



$$\begin{aligned} A &= L \times W \\ &= 4 \times 4 \\ &= 16 \times 6 \\ &= \underline{96\text{m}^2} \checkmark \end{aligned}$$

b)



$$\begin{aligned} A_1 &= \frac{1}{2} b \times h \\ &= \frac{1}{2} (8 \times 3) \\ &= 12 \times 2 \\ &= \underline{24\text{m}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_2 &= L \times W \\ &= 8.5 \times 6 \\ &= \underline{51\text{cm}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_3 &= L \times W \\ &= 6 \times 3 \\ &= \underline{18\text{cm}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_4 &= L \times W \\ &= 8 \times 6 \\ &= \underline{48\text{cm}^2} \checkmark \end{aligned}$$

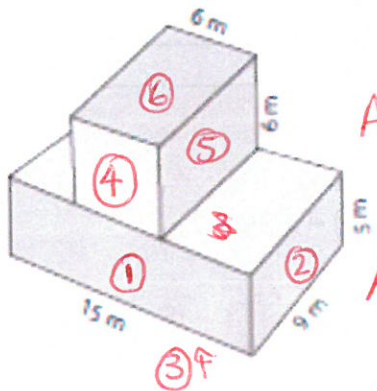
$$\text{TSA} = \underline{141\text{cm}^2} \checkmark$$

Question 10

8 marks

Calculate the surface area of the following composite solids

a)



$$\begin{aligned} A_1 &= L \times W \\ &= 15 \times 5 \\ &= \underline{75} \times 2 = \underline{150\text{m}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_2 &= L \times W \\ &= 9 \times 5 \\ &= \underline{45\text{m}^2} \times 2 = \underline{90\text{m}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_3 &= L \times W \\ &= 15 \times 9 \\ &= \underline{135} \times 2 = \underline{270\text{m}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_4 &= L \times W \\ &= 6 \times 6 \\ &= \underline{36} \times 2 = \underline{72\text{m}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_5 &= L \times W \\ &= 9 \times 6 \\ &= \underline{54} \times 2 \\ &= \underline{108\text{m}^2} \checkmark \end{aligned}$$

$$\begin{aligned} A_6 &= L \times W \\ &= 6 \times 9 \\ &= \underline{54\text{m}^2} \checkmark \end{aligned}$$

$$\begin{aligned} &= 270 + 90 + 150 + 72 + 108 + 54 \\ &= \underline{690\text{m}^2} \checkmark \end{aligned}$$

Question 11

3 marks

Calculate the surface area of a cylinder with base radius of 10 cm and a height of 26 cm.
(HINT: Draw a diagram and use the formula if needed)



$$\begin{aligned}
 SA &= 2\pi r^2 + 2\pi r \times h \\
 &= 2\pi(10)^2 + 2\pi(10)(26) \\
 &= 2261.95 \text{ cm}^2
 \end{aligned}$$

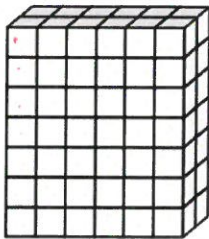
Volume

Question 12

2 marks

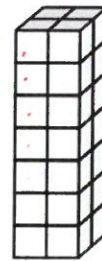
Count the cubes and calculate the volume of each prism. (Each cube is 1cm^3 .)

a)



$$\begin{aligned}
 \text{Volume} &= 6 \times 7 \times 2 \\
 &= 84 \text{ cm}^3
 \end{aligned}$$

b)



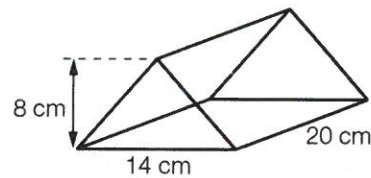
$$\begin{aligned}
 \text{Volume} &= 2 \times 7 \times 2 \\
 &= 28 \text{ cm}^3
 \end{aligned}$$

Question 13

2 marks

Work out the area of the triangle and use your answer to find the Volume of the shape.

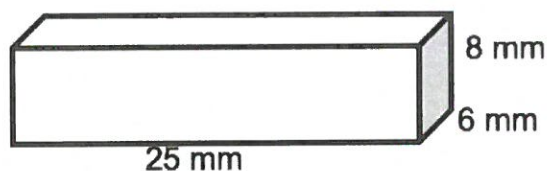
$$\begin{aligned}
 A &= \frac{1}{2} b \times h \\
 &= \frac{1}{2} (14) \times 8 \\
 &= 56 \text{ cm}^2 \times 20 \\
 &= 1120 \text{ cm}^3
 \end{aligned}$$



Question 14**4 marks**

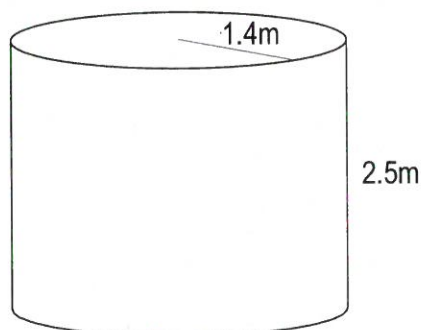
Calculate the Volume of the following 3D Shapes.

a)



$$\begin{aligned} V &= l \times w \times h \\ &= 25 \times 6 \times 8 \\ &= 1200 \text{ mm}^3 \end{aligned}$$

b)



$$\begin{aligned} V &= \pi r^2 \times h \\ &= \pi (1.4)^2 \times 2.5 \\ &= 15.39 \text{ m}^3 \end{aligned}$$

Question 15**3 marks**

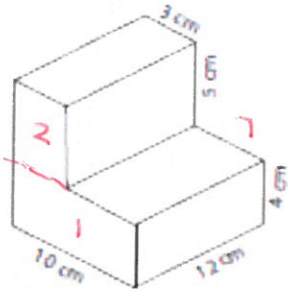
A cylinder has a volume of 375 cm^3 and a height of 15cm, find its radius.

$$\begin{aligned} V &= \pi r^2 \times h \\ 375 &= \pi r^2 \times 15 \\ r^2 &= 7.96 \\ r &= 2.82 \text{ cm} \end{aligned}$$

Question 16

4 marks

Calculate the volume of the following solid



$$\begin{aligned}
 V_1 &= L \times W \times H \\
 &= 10 \times 12 \times 4 \checkmark \\
 &= \underline{480 \text{ cm}^3} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 V_2 &= L \times W \times H \\
 &= 3 \times 5 \times 4 \checkmark \\
 &= \underline{60 \text{ cm}^3} \checkmark
 \end{aligned}$$

$$V_1 + V_2 = 540 \text{ cm}^3 \checkmark$$

END OF TEST

